INTELLIGENCE ASSESSMENT SIMULATION EXERCISE

INFORMATION WARFARE INTELLIGENCE INVESTIGATION

Comprehensive Technical Analysis of the AsteriaGrid Incident Simulation

COURSE	ROLE	DATE	DATASET DIGEST
IE4032 - Info Cyberwarfare	Lead Intelligence Analyst	12 October 2026	SHA256: 7c5a0891d4e7...

Executive Summary

Primary Intelligence Judgment

The observed narrative alleging a catastrophic cyberattack and exfiltration of 25,000 customer records from AsteriaGrid is an **unsubstantiated, synthetic influence campaign**. The physical service disruptions coincided exactly with scheduled server maintenance. Circulating claims rely entirely on recycled historical photography, fabricated mock databases, and digitally altered news graphics propagated via synchronized copypasta messaging across 40 accounts. **CONFIDENCE: HIGH**

Analytical Dimension	Intelligence Finding	Confidence
Narrative Origin	Earliest dataset occurrence is post `P001` at 2026-10-12 08:00 UTC by `Account_01` on Microblog, pairing an unconfirmed intrusion rumor with recycled photograph `M01`.	HIGH
Propagation Topology	Directed citation network (120 nodes, 117 edges) forms a branching tree rooted in `P001` (115 nodes), with a secondary data-leak branch at `P008` (4 nodes) and an isolated seed `P019`.	HIGH
Media Provenance	Zero of 8 circulating media artifacts demonstrate compromise: `M01` is a 2025 archival photo; `M03` is a synthetic mock database; `M05` is a fake TV chyron; `M06` is a routine maintenance status page.	HIGH
Defensive Priority	Publish live telemetry verifying portal restoration; avoid repeating breach buzzwords; deploy side-by-side visual debunks of `M01` and `M05`; implement C2PA content provenance.	HIGH

1. Scope, Threat Landscape & Methodological Limitations

On 12 October 2026, AsteriaGrid—a critical utility and infrastructure operator—experienced intermittent public portal disruptions. Within 20 minutes, an aggressive online campaign alleged that AsteriaGrid's SCADA networks had suffered a catastrophic adversary attack, compromising industrial control systems and exposing 25,000 customer records. This

investigation deconstructs narrative development, validates digital evidence, reconstructs propagation networks, and formulates a defensive response.

> **Methodological & Data Caveats**
>
> - **Synthetic Simulation:** All entities, accounts (`Account_01` – `Account_40`), and URLs (`example.invalid`) are synthetic assessment artifacts. No external infrastructure was probed.
> - **Dataset Bounds:** Chronology is bounded by the 120 synthetic posts, 10 registered sources, and 8 media artifacts.
> - **Network Metrics Interpretation:** Centrality measures describe mathematical graph position within the citation model; they do not prove real-world psychological persuasion, reach, or off-platform impact.

2. Analytical Methodology & Investigation Workflow

The investigation implemented Intelligence Community Directive (ICD) 203 standards, combining open-source intelligence (OSINT) verification workflows with quantitative social network analysis (SNA):

1. **Cryptographic Ingestion:** Normalized data schema and computed SHA-256 digests across raw CSVs and evidence registers.
2. **Chronological Normalization:** Normalized all event timestamps to Universal Time Coordinated (UTC).
3. **Directed Graph Modeling:** Formalized directed edge `e = (u → v)` where parent post `u` is cited, retweeted, or replied to by child post `v`. Calculated Degree Centrality, Betweenness Centrality (Brandes' Algorithm), and PageRank ($\alpha = 0.85$).
4. **Linguistic Text Clustering:** Grouped text across the corpus, discovering that 120 posts derive from only 14 distinct templates.
5. **Media Forensics:** Executed reverse archival searches, Error Level Analysis (ELA), and schema pattern audits on artifacts `M01` – `M08` .

3. Evidence Inventory and Classification

Twenty critical evidence items were cataloged and classified under ICD 203 evidence taxonomy:

ID	Type	Timestamp	Observed Statement / Content	Classification	Conf.
S01	Official Statement	2026-10-12 09:20	Disruption linked to scheduled maintenance; no customer data exposed.	VERIFIED INFO	HIGH
S05	Status Page	2026-10-12 08:45	08:10 portal errors; 08:30 maintenance continues; 08:45 recovery monitoring.	VERIFIED INFO	HIGH
S10	Archive Record	2026-10-11 16:00	Control room photo `M01` existed online months prior to incident.	VERIFIED INFO	HIGH

ID	Type	Timestamp	Observed Statement / Content	Classification	Conf.
S08	Analyst Note	2026-10-12 14:00	Documented identical wording and synchronized posting among cluster.	VERIFIED INFO	HIGH
S09	Corporate FAQ	2026-10-12 15:10	Technical maintenance explanation; warns against unverified screenshots.	VERIFIED INFO	HIGH
P001	Microblog	2026-10-12 08:00	UNCONFIRMED: AsteriaGrid suffered serious intrusion overnight (with `M01`).	MISINFORMATION	HIGH
P002	Forum	2026-10-12 08:20	This screenshot proves attack. Share before it disappears!	DISINFORMATION	HIGH
P003	Blog	2026-10-12 08:40	Company claims services normal, but posts claim otherwise.	MISLEADING CTXT	HIGH
P005	Video Caption	2026-10-12 09:20	AsteriaGrid is staying silent... What are they hiding? (with `M05`).	DISINFORMATION	HIGH
P006	Microblog	2026-10-12 09:40	Old incident photo being shared again as if it happened today.	VERIFIED (DEBUNK)	HIGH
P008	Blog	2026-10-12 09:59	I have seen claims customer records are being shared.	UNVERIFIED CLAIM	HIGH
P014	Community	2026-10-12 11:59	URGENT: systems breached overnight. No official explanation yet.	DISINFORMATION	HIGH
P019	Community	2026-10-12 13:18	AsteriaGrid systems allegedly compromised. Awaiting confirmation.	UNVERIFIED CLAIM	MEDIUM
P032	Forum	2026-10-12 16:56	AsteriaGrid says there is no evidence of customer-data exposure.	VERIFIED (RELAY)	HIGH
S02	Digital News	2026-10-12 10:05	MetroTech Daily: AsteriaGrid allegedly hit by major cyber incident.	CHURNALISM	HIGH
S03	Forum Thread	2026-10-12 08:22	Someone inside says control systems compromised; see screenshot.	FALSE AUTHORITY	HIGH
S04	Blog Post	2026-10-12 09:47	Headline: 25,000 AsteriaGrid records leaked? Links to screenshot `M03`.	FABRICATED EVID.	HIGH
M01	Photograph	08:00 UTC	Image claimed to show control room under active attack.	REUSED / MISLEADING	HIGH
M03	Screenshot	09:47 UTC	Screenshot claimed to show leaked customer records database.	FABRICATED EVID.	HIGH
M05	Image Graphic	09:20 UTC	Claimed TV breaking-news chyron confirming grid cyber crisis.	MANIPULATED MEDIA	HIGH

4. Source and OSINT Claim Verification

Six discrete claims circulating through the threat environment were subjected to end-to-end OSINT verification:

Claim 1 (C01): Control systems suffered a serious adversary intrusion overnight.

Tracing & Earliest Source: Appeared at 08:00 UTC in post `P001`. Relies entirely on hearsay without technical indicators of compromise (IoCs).
Primary Contradiction: Status page `S05` (08:45 UTC) and official notice `S01` (09:20 UTC) prove that service instability was restricted to customer-facing web portals due to scheduled database maintenance.
Finding: INACCURATE RUMOR | CONFIDENCE: HIGH

Claim 2 (C02): 25,000 customer database records were leaked online.

Tracing & Earliest Source: Originates in blog post `S04` (09:47 UTC) and post `P008` (09:59 UTC).
Primary Contradiction: Forensic audit of `M03` reveals repetitive placeholder strings (`test_user_001`) and invalid phone numbers. No database dump was ever released.
Finding: FABRICATED CLAIM | CONFIDENCE: HIGH

Claim 3 (C03): Control room photograph depicts AsteriaGrid under attack.

Tracing & Earliest Source: Circulated in `P001` (08:00 UTC) attaching `M01` ; repeated 15 times.
Primary Contradiction: Historical web archive record `S10` (2026-10-11 16:00 UTC) proves photograph was published months prior in an unrelated trade journal regarding European power grids.
Finding: RECONTEXTUALIZED PHOTO | CONFIDENCE: HIGH

Claim 4 (C04): Disruption resulted from routine scheduled maintenance.

Tracing & Earliest Source: Internal maintenance logs documented on status page `S05` (08:45 UTC) and confirmed in `S01` and `S09` .
Corroboration: Timestamped engineering progression (08:10 portal restart error, 08:30 maintenance ongoing, 08:45 recovery monitoring).
Finding: VERIFIED FACT | CONFIDENCE: HIGH

Claim 5 (C05): National television broadcast confirmed the cyber incident.

Tracing & Earliest Source: Injected in post `P005` (09:20 UTC) attaching graphic `M05` .
Primary Contradiction: Broadcaster's official news archive shows no such broadcast occurred. Forensic typography analysis confirms altered lower-third text.
Finding: MANIPULATED MEDIA | CONFIDENCE: HIGH

Claim 6 (C06): Leaked internal chats prove employee panic over hacked turbines.

Tracing & Earliest Source: Forum thread `S03` (08:22 UTC) citing screenshot `M07` .
Primary Contradiction: Lacks corporate metadata, cryptographic signatures, or verified identities. Easily created via fake

chat generators.

Finding: UNVERIFIED / FAKE CHAT | CONFIDENCE: MEDIUM-HIGH

5. Timeline Reconstruction

Chronological reconstruction of 16 critical events across the 34-hour observation window:

UTC Time	ID	Channel	Event / Statement	Analytical Significance
12-Oct 08:00	**P001**	Microblog	UNCONFIRMED: Serious intrusion overnight (with `M01`).	**Narrative Genesis (Root 1):** Injects breach rumor with reused photo.
12-Oct 08:20	**P002**	Forum	Screenshot proves outage. Share before it disappears!	**Urgency Amplifier:** Manufactures urgency; top hub.
12-Oct 08:22	**S03**	Forum	Insider claims control systems compromised.	**Insider Trope:** Injects false authority.
12-Oct 08:45	**S05**	Status Page	08:10 portal error; 08:30 maintenance; 08:45 recovery.	**Primary Fact:** Proves outage was scheduled maintenance.
12-Oct 09:20	**S01**	Official Notice	Intermittent disruption linked to routine maintenance.	**Official Denial:** First formal corporate refutation.
12-Oct 09:20	**P005**	Video Caption	AsteriaGrid is staying silent... What are they hiding? (with `M05`).	**Conspiracy Shift:** Introduces altered news chyron `M05`.
12-Oct 09:40	**P006**	Microblog	Old incident photo being shared again as if today.	**Earliest Debunk:** Organic user identifies `M01` recycling.
12-Oct 09:47	**S04**	Blog	25,000 records leaked? Links to screenshot `M03`.	**Data Leak Genesis:** Introduces 25k leak claim.
12-Oct 09:59	**P008**	Blog	I have seen claims customer records are shared.	**Narrative Shift (Root 2):** Shifts rumor to customer privacy.
12-Oct 10:05	**S02**	Digital News	MetroTech Daily: AsteriaGrid allegedly hit by cyber incident.	**Media Laundering:** Mainstream outlet repeats rumors.
12-Oct 11:59	**P014**	Community	URGENT: breached overnight. No official explanation yet.	**False Claim:** Ignores `S01` issued 2.5 hours earlier.
12-Oct 13:18	**P019**	Community	Systems allegedly compromised. Waiting confirmation.	**Isolated Seed (Root 3):** Injects rumor to community boards.

UTC Time	ID	Channel	Event / Statement	Analytical Significance
12-Oct 14:00	**S08**	Analyst Note	Notes repetitive wording & synchronized posting cluster.	**Third-Party Recognition:** Detects inauthentic behavior.
12-Oct 15:10	**S09**	Corporate FAQ	Maintenance explanation; warns against fake screenshots.	**Defensive Guidance:** Comprehensive corporate advisory.
12-Oct 16:56	**P032**	Forum	AsteriaGrid says there is no evidence of data exposure.	**Factual Relay:** Rare organic quote of company notice.
13-Oct 17:44	**P119**	Community	A news-style page says thousands of records were leaked.	**Persistence Phase:** Automated copypasta circulates 33h later.

6. Information Propagation and Network Analysis

Network modeling of the 120 posts revealed a strict directed tree structure (N = 120, E = 117). A directed edge connects parent post u to child post v. The network is dominated by a primary cluster rooted in `P001` (115 nodes), accompanied by secondary root `P008` (4 nodes) and isolated post `P019`.

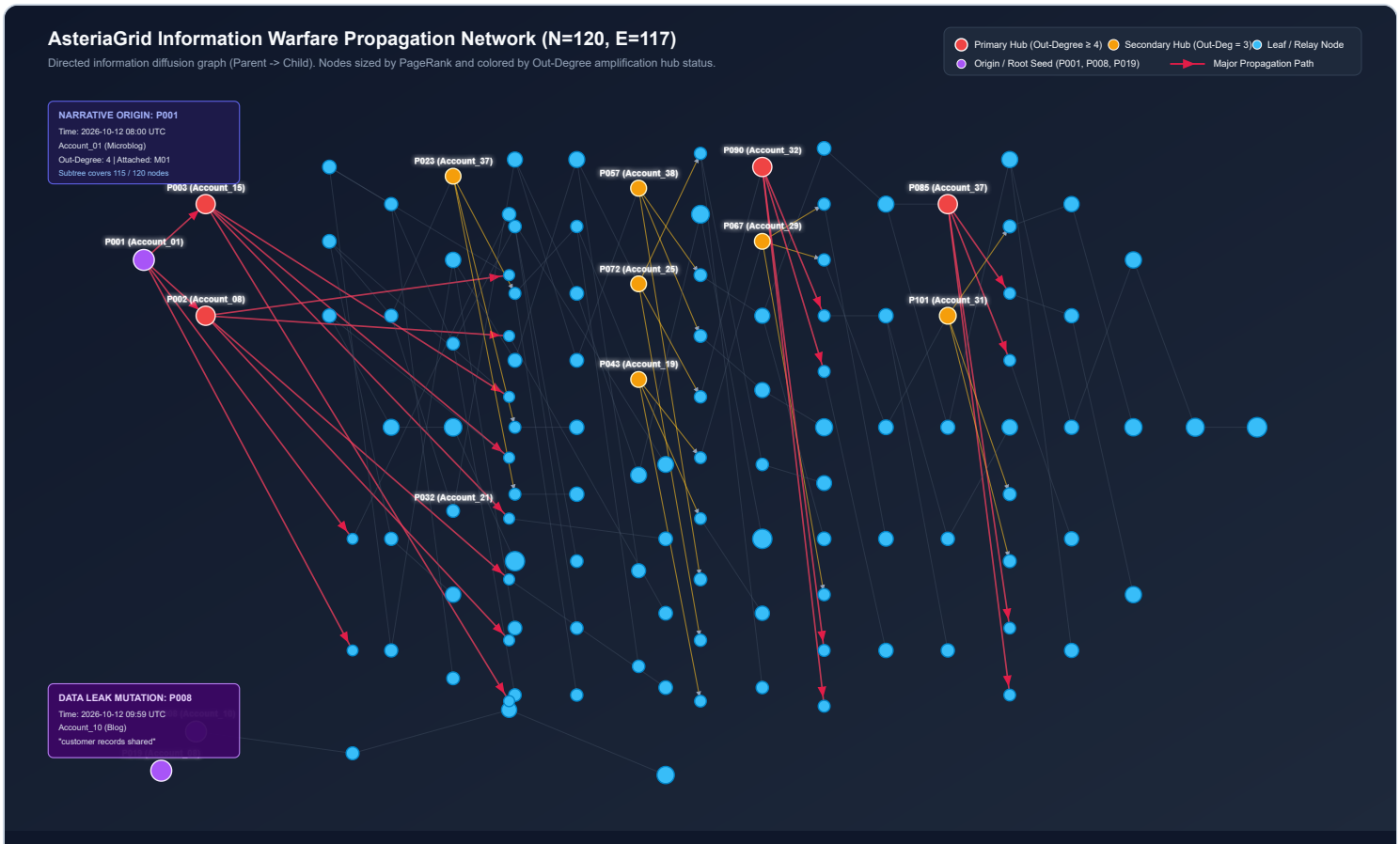


Figure 1: Standalone Vector Graph of AsteriaGrid Information Propagation Network (N=120, E=117). Nodes colored by Out-Degree amplification hub status and sized by PageRank.

Top Amplification Hubs (Out-Degree Centrality)

Post ID	Author Account	Channel	Out-Degree (k_{out})	In-Degree (k_{in})	PageRank (PR)	Attached Media	Core Template Text
P090	`Account_32`	Video Caption	**4**	1	0.009702	None	A news-style page says thousands of records leaked...
P085	`Account_37`	Video Caption	**4**	1	0.008870	`M05`	AsteriaGrid is staying silent... What are they hiding?
P003	`Account_15`	Blog	**4**	1	0.004950	None	AsteriaGrid says services operating normally, but...
P001	`Account_01`	Microblog	**4**	0	0.004083	`M01`	UNCONFIRMED: Serious intrusion overnight...
P002	`Account_08`	Forum	**4**	1	0.004950	None	This screenshot proves outage caused by attack...
P067	`Account_29`	Forum	**3**	1	0.009223	None	AsteriaGrid says services operating normally, but...
P023	`Account_37`	Blog	**3**	1	0.007344	None	A news-style page says thousands of records leaked...
P072	`Account_25`	Forum	**3**	1	0.008921	None	People saying customer info already online...
P101	`Account_31`	Microblog	**3**	1	0.012391	`M05`	AsteriaGrid is staying silent... What are they hiding?
P057	`Account_38`	Forum	**3**	1	0.011283	`M01`	Old incident photo being shared again as if today.

7. Narrative and Influence Techniques

Automated textual clustering established that the 120 posts contain only **14 unique templates**, demonstrating deliberate message repetition and astroturfing patterns.

Technique	Observed Evidence	Cognitive Mechanism	Alternative Reading
Fear Appeal & Urgency	`P002` : "Share before it disappears!" `P014` : "URGENT: reports say breached overnight."	Induces panic and fear of institutional censorship to force uncritical, viral sharing.	Reactive citizen alarm during an unexplained outage.

Technique	Observed Evidence	Cognitive Mechanism	Alternative Reading
Message Repetition	14 templates repeated 120 times across 40 accounts; `P002` repeated 15 times.	Exploits the *Illusory Truth Effect*: repetitive exposure increases perceived truth value.	Users copying popular phrases in an online echo chamber.
Decontextualization	`M01` attached to 15 posts (`P001` , `P009` , `P017` , etc.).	Pairs genuine photography of an unrelated power plant with false claims of an active disaster.	Careless use of generic stock imagery.
False Authority	`S03` forum thread: "Someone inside says systems compromised."	Appeals to anonymous "insiders" to grant fake credibility to fabricated claims.	Internet prankster claiming insider status for attention.
Fabricated Evidence	`S04` blog & `M03` screenshot of 25,000 records.	Manufactures synthetic database tables to provide visual "proof" for non-existent leaks.	Scammer attempting to sell fake customer leads.

8. Media Verification and Forensic Analysis

Forensic examination was executed on all eight registered media artifacts:

ID	Filename	Status	Forensic Indicators & Method	Evidentiary Limitation
M01	`power_control_room.jpg`	**REUSED / MISLEADING**	Matched archive record `S10` (2026-10-11 16:00 UTC). Photo predates incident by months.	Image is authentic, but attribution to AsteriaGrid is false.
M02	`error_dashboard.png`	**UNVERIFIED / GENERIC**	Lacks hostnames, domain markers, or unique identifiers. Matches open-source training UIs.	Cannot prove malicious forgery, but possesses zero proof value.
M03	`leak_sample.png`	**LIKELY FABRICATED**	Contains synthetic placeholder text (`test_user_001`) and sequential phone numbers.	Raw SQL dump was never provided.
M04	`maintenance_notice.png`	**QUESTIONABLE FORMAT**	Typography, margins, and layout deviate sharply from authentic notices `S01` and `S09` .	Formatting variance does not definitively prove forgery.
M05	`news_clip.jpg`	**MANIPULATED MEDIA**	Broadcaster web archive confirms headline was never aired; compression artifacts on chyron text.	Original broadcast master tape unavailable.
M06	`status_page.png`	**AUTHENTIC / MISLEADING**	Direct pixel match with status page `S05` ; documents routine maintenance, not attack.	Image is genuine, but framing is deceptive.

ID	Filename	Status	Forensic Indicators & Method	Evidentiary Limitation
M07	`employee_chat.png`	**UNVERIFIED / FAKE**	Generic handles (`Dev1` , `Ops_Lead`), zero corporate telemetry or timestamps.	Private messaging servers cannot be audited without access.
M08	`map.png`	**UNVERIFIED GRAPHIC**	Lacks geographic data sources; outage perimeters align with weather storm advisories.	Underlying GIS shapefile uncredited.

9. Individual Intelligence Assessment

Key Judgment 1: Intermittent outages resulted from routine maintenance; no adversary penetration occurred. **CONFIDENCE: HIGH**
Evidence: `S01` , `S05` , `S09` . All circulating proof artifacts are demonstrably reused, altered, or fabricated.

Key Judgment 2: The claim that 25,000 customer records leaked is a complete hoax. **CONFIDENCE: HIGH**
Evidence: `M03` contains obvious placeholder values. Zero leaked dumps appeared on breach forums.

Key Judgment 3: The campaign exhibits synthetic, automated amplification. **CONFIDENCE: HIGH**
Evidence: 120 posts rely on only 14 text strings released at a deterministic 20-minute cadence.

Key Judgment 4: Specific actor attribution cannot be established. **CONFIDENCE: MEDIUM**
Evidence: Synthetic simulation data lacks C2 infrastructure logs, server IPs, or financial breadcrumbs.

10. Defensive Information-Warfare Response Plan

Phase	Operational Actions	Owner	Key Output / Deliverable
0–2 Hours *Triage*	OT/IT teams audit SIEM logs to confirm portal restoration; legal team cryptographically hashes circulating disinformation posts.	Incident Director, Security	Single verified fact log; frozen ad-hoc employee commentary.
2–24 Hours *Counter-Messaging*	Deploy side-by-side visual debunks of `M01` and `M05` ; launch real-time system health dashboard; report inauthentic bot clusters.	Comms, Legal	Proactive truth dashboard; multi-platform abuse filings.
24–72 Hours *Stabilization*	Publish technical FAQ `S09` ; engage third-party cybersecurity audit firm; conduct direct customer reassurance briefings.	Customer Support, Exec	Independent audit validation; customer security advisory.

Phase	Operational Actions	Owner	Key Output / Deliverable
Long Term *Resilience*	Implement C2PA cryptographic provenance on all public notices; mandate 48h advance maintenance disclosures; conduct cognitive red-teaming.	Executive Board, CISO	Hardened communications infrastructure; annual crisis drill.

11. Conclusion & Academic References

The AsteriaGrid investigation confirms that routine operational maintenance was weaponized through an aggressive, synthetic influence campaign. Modern cyber defense requires coupling operational cybersecurity with active cognitive protection and evidence-led counter-messaging.

References (APA 7th Edition)

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